Mathematical Methods I

Circulation, Density, and Orientation

Gabriel Velázquez

1. Circulation Along a Loop

Let $\mathbf{V}$ be a vector field: the velocity of a fluid or a force. We walk along a closed curve C and, at each point, ask whether the field *pushes along* or *against* the direction in which we are walking. Only the component of $\mathbf{V}$ *tangent* to the curve matters; the perpendicular component does not contribute to the trip.

Definition 1.1 (Circulation)

Let C be an oriented closed curve, parametrized by $\mathbf{r}(t) = (x(t), y(t), z(t))$ with $a \le t \le b$, and let $\mathbf{V} = (P, Q, R)$ be a continuous field. The **circulation** of $\mathbf{V}$ along C is

$$\oint_C \mathbf{V} \cdot d\mathbf{r} = \oint_C V_T \, ds = \int_a^b \mathbf{V}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt, \qquad V_T = \mathbf{V} \cdot \mathbf{T}, \tag{1}$$

where $\mathbf{T}$ is the unit tangent vector. With the 1-form $\alpha = P\,dx + Q\,dy + R\,dz$ associated with the field, this is written

$$\oint_C \alpha = \int_a^b (P\,x' + Q\,y' + R\,z') \, dt. \tag{2}$$

The 1-form α acts as a machine that *consumes* the velocity vector $\mathbf{r}'(t)$ of the trip and returns a number: $\alpha_p(\mathbf{w}) = \mathbf{V}(p) \cdot \mathbf{w}$. The integral adds up those numbers along the curve. Note that the integral only needs α and the curve; it does not require a metric to measure lengths.

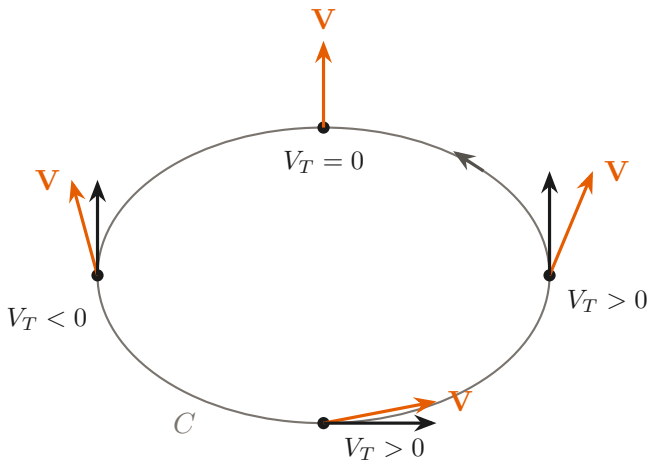


Figure 1: The curve is traversed counterclockwise. The orange arrows are the field $\mathbf{V}$; the black ones are its projection V_T onto the direction of travel. Where the field helps the trip it contributes a $+$ sign; where it opposes the trip, a $-$ sign; where it is perpendicular, it contributes nothing.

Two Physical Readings of the Same Integral

- **Fluids:** if $\mathbf{V}$ is the velocity, the circulation measures the overall tendency of the fluid to rotate around C. A paddle wheel placed in the flow turns when the velocity is greater on one side than on the other.
- **Forces:** if $\mathbf{V} = \mathbf{F}$ is a force field, the circulation is the total work done in carrying a particle around the closed path.

Reversing the orientation changes the sign: $\oint_{-C} \alpha = -\oint_C \alpha$.

Example 1.2 (Two Fields on the Unit Circle)

Let C: $\mathbf{r}(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, with $\mathbf{r}'(t) = (-\sin t, \cos t)$.

- **Rotation field** $\mathbf{V} = (-y, x)$, that is, $\alpha = -y\,dx + x\,dy$. Here $\mathbf{V}(\mathbf{r}(t)) = (-\sin t, \cos t)$ coincides with $\mathbf{r}'(t)$, so
$$\oint_C \alpha = \int_0^{2\pi} (\sin^2 t + \cos^2 t)\,dt = 2\pi.$$
Traversed backwards, the circulation is -2π.
- **Radial field** $\mathbf{V} = (x, y)$, that is, $\alpha = x\,dx + y\,dy$. Now $\mathbf{V} \cdot \mathbf{r}' = -\cos t \sin t + \sin t \cos t = 0$ at every point: the field is perpendicular to the curve and the circulation is 0.

2. How to Compute It: the Square

For a curve with corners, the boundary is divided into smooth pieces and the integrals over each piece are added. Consider the square $Q = [0, a] \times [0, a]$ traversed counterclockwise and the 1-form $\alpha = P(x, y)\,dx + Q(x, y)\,dy$.

Proposition 2.1 (Circulation Around a Square)

The boundary ∂Q consists of four pieces:

$$\begin{aligned}
&C_1 \text{ (bottom):} && y = 0,\ dy = 0 && x : 0 \to a && \int_0^a P(x,0)\,dx \\
&C_2 \text{ (right):} && x = a,\ dx = 0 && y : 0 \to a && \int_0^a Q(a,y)\,dy \\
&C_3 \text{ (top):} && y = a,\ dy = 0 && x : a \to 0 && -\int_0^a P(x,a)\,dx \\
&C_4 \text{ (left):} && x = 0,\ dx = 0 && y : a \to 0 && -\int_0^a Q(0,y)\,dy
\end{aligned}$$

and therefore

$$\oint_{\partial Q} \alpha = \int_0^a [P(x,0) - P(x,a)]\,dx + \int_0^a [Q(a,y) - Q(0,y)]\,dy. \tag{3}$$

The negative sign on C_3 and C_4 appears because they are traversed against the direction of the axes. This is why opposite sides are *subtracted*: what matters is how much P changes from bottom to top and how much Q changes from left to right. This observation is the seed of the circulation density in the next section.

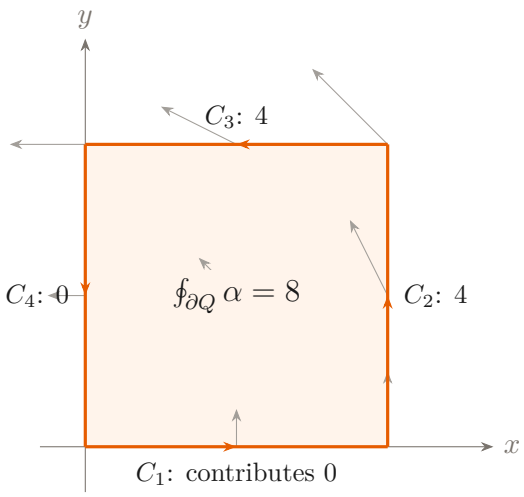


Figure 2: Square $[0,2]^2$ with the field $(-y, x)$ from Example 2.2 (gray arrows, drawn at reduced scale). The bottom and left sides are *perpendicular* to the field and contribute nothing; the right and top sides go along the field and contribute 4 each.

Example 2.2 (Circulation of $-y\,dx + x\,dy$ on $[0,2]^2$)

With $a = 2$:

- C_1: $y = 0$, so $\alpha = 0$ and the integral is 0.
- C_2: $x = 2$, $dx = 0$: $\displaystyle\int_0^2 2\,dy = 4$.
- C_3: $y = 2$, $dy = 0$, x from 2 to 0: $\displaystyle\int_2^0 (-2)\,dx = 4$.
- C_4: $x = 0$, so $\alpha = 0$ and the integral is 0.

Total: $\oint_{\partial Q} \alpha = 0+4+4+0 = \mathbf{8}$. Since the area of the square is 4, the result is exactly $2\times$ area. This is no coincidence; it is explained in Section 3.

3. Shrinking the Loop: Circulation Density

The circulation depends on the chosen loop. To know what the field does *at a point*, we shrink the loop around it and normalize by the enclosed area.

Definition 3.1 (Circulation Density)

Let p be a point, and let $\partial(\Delta A)$ be the counterclockwise boundary of a small region of area ΔA containing p. The **circulation density** of α at p is

$$\lim_{\Delta A \to 0} \frac{1}{\Delta A} \oint_{\partial(\Delta A)} \alpha. \tag{4}$$

It is the local rate of rotation of the field: the circulation *per unit area*.

To compute it we take an infinitesimal rectangle $[x, x+\Delta x] \times [y, y+\Delta y]$ and approximate each side by the value of the integrand at its starting point.

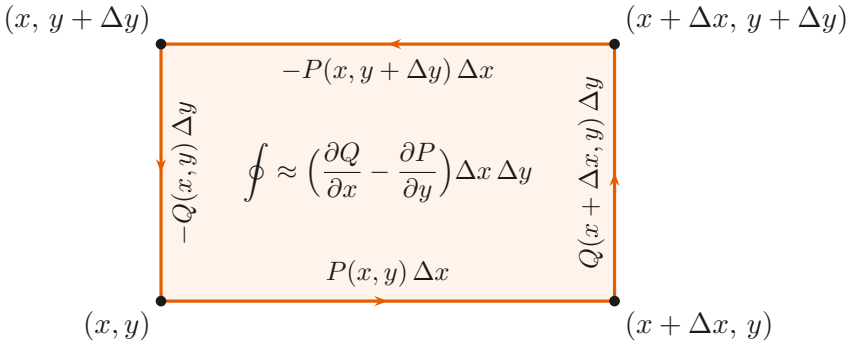


Figure 3: Contribution of each side of the infinitesimal rectangle. On the top and bottom, the values of P at two heights are compared; on the left and right, the values of Q at two positions.

Adding the four sides:

$$\begin{aligned}\oint_{\partial(\Delta A)} \alpha &\approx P(x,y)\Delta x + Q(x+\Delta x,y)\Delta y - P(x,y+\Delta y)\Delta x - Q(x,y)\Delta y \\ &= -\big[P(x,y+\Delta y) - P(x,y)\big]\Delta x + \big[Q(x+\Delta x,y) - Q(x,y)\big]\Delta y \\ &\approx \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big)\Delta x\,\Delta y.\end{aligned}$$

Theorem 3.2 (Circulation Density in the Plane)

If $\alpha = P\,dx + Q\,dy$ is of class C^1, its circulation density at (x, y) is

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}. \tag{5}$$

How to Read the Sign

- $\partial Q/\partial x > 0$: the vertical component of the field grows toward the right. The right side (which we traverse upward) receives more push than the left side (which we traverse downward): counterclockwise rotation, contribution $+$.
- $\partial P/\partial y > 0$: the horizontal component grows upward. On the top side we move to the left, against the push, which is stronger there than at the bottom: clockwise rotation, contribution $-$.

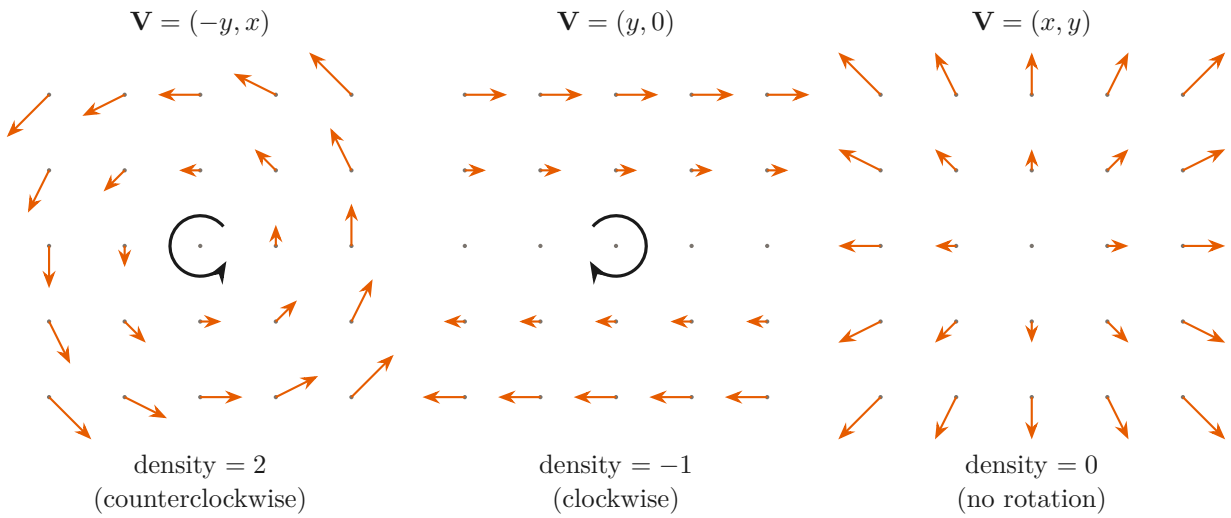


Figure 4: Three fields with constant circulation density. A paddle wheel placed at any point turns counterclockwise in the first, turns clockwise in the second, and does not turn in the third.

Example 3.3 (Three Fields, Three Behaviors)

- $\alpha = -y\,dx + x\,dy$: $\partial_x Q - \partial_y P = 1 - (-1) = 2$. This is a rigid rotation with angular velocity 1. It explains the result of Example 2.2: circulation $= 2 \times$ area $= 2 \times 4 = 8$. In general, the circulation density of a fluid is twice its local angular velocity.
- $\alpha = y\,dx$: $\partial_x Q - \partial_y P = 0 - 1 = -1$. *The streamlines are straight*, and yet there is rotation: the paddle wheel turns because the velocity changes from one layer to the next.
- $\alpha = x\,dx + y\,dy$: $\partial_x Q - \partial_y P = 0 - 0 = 0$. A radial field expands the fluid but does not make it rotate.

4. Oriented Area and the Wedge Product

The circulation depends on the direction of travel (counterclockwise or clockwise). To rebuild it as *density* $\times$ *area* we need a notion of area that also carries a direction: a signed area. That is a 2-form.

Definition 4.1 (The 2-Form $dx \wedge dy$)

$dx \wedge dy$ is a function that takes *two* vectors $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ and returns the oriented area of the parallelogram they span:

$$(dx \wedge dy)(\mathbf{u}, \mathbf{v}) = u_1 v_2 - u_2 v_1 = \det \begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}. \tag{6}$$

The sign is $+$ if going from $\mathbf{u}$ to $\mathbf{v}$ through the smaller angle is counterclockwise, and $-$ if it is clockwise.



$(dx \wedge dy)(\mathbf{u}, \mathbf{v}) = +2$ counterclockwise

$(dx \wedge dy)(\mathbf{v}, \mathbf{u}) = -2$ clockwise

Figure 5: With $\mathbf{u} = (2, 0)$ and $\mathbf{v} = (1, 1)$ the parallelogram has area 2 in both cases. What changes is the *order* of the vectors and, with it, the direction in which the boundary is traversed. Swapping the order flips the sign.

Proposition 4.2 (Properties of the Wedge Product)

For 1-forms ω, η and scalars a, b:

$$dx \wedge dy = -\,dy \wedge dx, \qquad dx \wedge dx = 0, \qquad \omega \wedge (a\eta) = a\,(\omega \wedge \eta). \tag{7}$$

The second identity says that two parallel vectors span a degenerate parallelogram, of area 0.

In $\mathbb{R}^3$: Three Planes, Three Areas

A parallelogram in space has three *shadows* on the coordinate planes, and each shadow is an oriented area:

$$(dy \wedge dz)(\mathbf{u}, \mathbf{v}) = u_2v_3 - u_3v_2, \quad (dz \wedge dx)(\mathbf{u}, \mathbf{v}) = u_3v_1 - u_1v_3, \quad (dx \wedge dy)(\mathbf{u}, \mathbf{v}) = u_1v_2 - u_2v_1. \tag{8}$$

These three quantities are exactly the components of the cross product $\mathbf{u} \times \mathbf{v}$: the normal vector whose length is the area and whose direction gives the orientation.

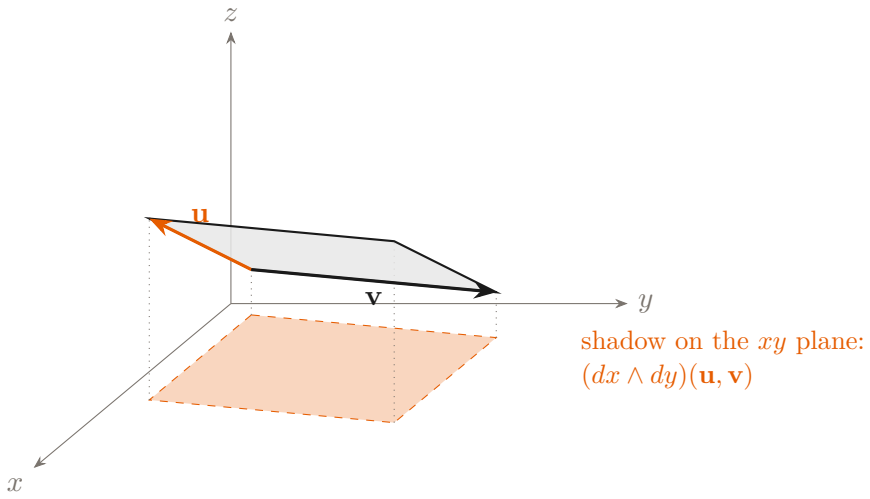


Figure 6: A parallelogram in $\mathbb{R}^3$ and its shadow on the xy plane (orange). Evaluating $dx \wedge dy$ at $(\mathbf{u}, \mathbf{v})$ measures the oriented area of the shadow. Likewise, $dy \wedge dz$ and $dz \wedge dx$ measure the shadows on the other two planes.

Example 4.3 (The Three Shadows)

Let $\mathbf{u} = (1.5,\, 0,\, 1.2)$ and $\mathbf{v} = (0.4,\, 2.4,\, 0)$ be the vectors in Figure 6. Then

$$\begin{aligned}(dy \wedge dz)(\mathbf{u}, \mathbf{v}) &= 0(0) - 1.2(2.4) = -2.88,\\ (dz \wedge dx)(\mathbf{u}, \mathbf{v}) &= 1.2(0.4) - 1.5(0) = 0.48,\\ (dx \wedge dy)(\mathbf{u}, \mathbf{v}) &= 1.5(2.4) - 0(0.4) = 3.6,\end{aligned}$$

and indeed $\mathbf{u} \times \mathbf{v} = (-2.88,\, 0.48,\, 3.6)$. The shadow on the xy plane is the largest, and the one on the zx plane is very small: the normal vector points mostly along z, and the negative sign of the first component indicates that the shadow on the yz plane is oppositely oriented.

5. The Exterior Derivative: Density × Oriented Area

We now have the two ingredients: the circulation density (Section 3) and the oriented area (Section 4). The exterior derivative d combines them into a single object.

Definition 5.1 (Exterior Derivative of a 1-Form)

For $\alpha = P\,dx + Q\,dy + R\,dz$ the 2-form $d\alpha$ is defined by

$$d\alpha = \Big(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\Big) dy \wedge dz + \Big(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\Big) dz \wedge dx + \Big(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\Big) dx \wedge dy. \tag{9}$$

In the plane, $d(P\,dx + Q\,dy) = (\partial_x Q - \partial_y P)\, dx \wedge dy$.

Each term has the form

$$\underbrace{(\text{circulation density in a coordinate plane})}_{\text{local curl}} \times \underbrace{(\text{oriented area projected onto that plane})}_{\text{basic 2-form}}.$$

The three coefficients are the components of the curl $\nabla \times \mathbf{V}$.

Theorem 5.2 (Geometric Meaning of $d\alpha$)

Let p be a point and $\mathbf{u}, \mathbf{v}$ two vectors. If Π_ε is the parallelogram with edges $\varepsilon\mathbf{u}$ and $\varepsilon\mathbf{v}$ and a vertex at p, traversed from $\mathbf{u}$ toward $\mathbf{v}$, then

$$\oint_{\partial \Pi_\varepsilon} \alpha = \varepsilon^2\, (d\alpha)_p(\mathbf{u}, \mathbf{v}) + o(\varepsilon^2), \qquad (d\alpha)_p(\mathbf{u}, \mathbf{v}) = (\nabla \times \mathbf{V})(p) \cdot (\mathbf{u} \times \mathbf{v}). \tag{10}$$

In other words, *the circulation around a small parallelogram is the curl times its oriented area.* The second equality follows from comparing the coefficients of $d\alpha$ with the components of $\mathbf{u} \times \mathbf{v}$ from Example 4.3. In the planar case with $\mathbf{u} = \Delta x\, \mathbf{e}_1$ and $\mathbf{v} = \Delta y\, \mathbf{e}_2$ it recovers the rectangle computation of Section 3.

Example 5.3 (Computing $d\alpha$)

- **Planar rotation.** For $\alpha = -y\,dx + x\,dy$, $d\alpha = (1-(-1))dx \wedge dy = 2\,dx \wedge dy$.
- **In space.** For $\alpha = z\,dx + x\,dy + y\,dz$ we have $P = z$, $Q = x$, $R = y$, so

$$d\alpha = (1-0)\,dy \wedge dz + (1-0)\,dz \wedge dx + (1-0)\,dx \wedge dy, \qquad \nabla \times \mathbf{V} = (1,1,1).$$

6. Oriented Volume and the 3-Form $dx \wedge dy \wedge dz$

Just as $dx \wedge dy$ measures the signed area of a parallelogram, in $\mathbb{R}^3$ there is an analogous object for three vectors: the *signed volume* of a parallelepiped.

Definition 6.1 (Oriented Volume)

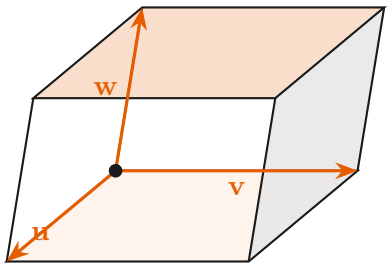
The 3-form $dx \wedge dy \wedge dz$ takes three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ and returns

$$(dx \wedge dy \wedge dz)(\mathbf{u}, \mathbf{v}, \mathbf{w}) = \det \begin{pmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{pmatrix} = \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}). \tag{11}$$

Its absolute value is the volume of the parallelepiped with edges $\mathbf{u}, \mathbf{v}, \mathbf{w}$. The sign is $+$ if the triple $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ follows the **right-hand rule** (thumb, index, and middle finger), and $-$ otherwise.

The orientation of an area was read as counterclockwise or clockwise; that of a volume is read as right-handed or left-handed. The sign rules are the same as in Section 4:

- Swapping *any two* vectors flips the sign: $dx \wedge dy \wedge dz = -\,dy \wedge dx \wedge dz$.
- If the three vectors are coplanar, the parallelepiped is flat and the volume is 0. In particular, any wedge product with a repeated factor vanishes: $dx \wedge dx \wedge dz = 0$.



$(dx \wedge dy \wedge dz)(\mathbf{u}, \mathbf{v}, \mathbf{w}) = +4.5$
right-handed triple

Figure 7: Parallelepiped spanned by $\mathbf{u} = (1.5, 0, 0)$, $\mathbf{v} = (0, 2, 0)$, and $\mathbf{w} = (0.3, 0.4, 1.5)$. The value $+4.5$ is its volume, with a positive sign because $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is right-handed. With $(\mathbf{v}, \mathbf{u}, \mathbf{w})$ one would obtain -4.5.

Example 6.2 (Computing Oriented Volumes)

With the vectors of Figure 7, $\mathbf{v} \times \mathbf{w} = (3,\ 0,\ -0.6)$, so

$$(dx \wedge dy \wedge dz)(\mathbf{u}, \mathbf{v}, \mathbf{w}) = 1.5(3) = 4.5.$$

Swapping the first two vectors gives -4.5. And if $\mathbf{w}$ is replaced by $\mathbf{u} + \mathbf{v}$ (which lies in the plane of $\mathbf{u}$ and $\mathbf{v}$), the determinant vanishes.

A 2-Form Also Measures Something: Flux

A 2-form $\beta = A\,dy \wedge dz + B\,dz \wedge dx + C\,dx \wedge dy$ evaluated on two vectors gives, by Section 4,

$$\beta(\mathbf{u}, \mathbf{v}) = (A, B, C) \cdot (\mathbf{u} \times \mathbf{v}). \tag{12}$$

If $\mathbf{V} = (A, B, C)$, this is the *flux* of $\mathbf{V}$ through the parallelogram with edges $\mathbf{u}, \mathbf{v}$: the component of the field perpendicular to the patch, times its oriented area. The 1-form measured how much the field pushes *along* a curve; the 2-form measures how much it passes *through* a surface.

Proposition 6.3 (Exterior Derivative of a 2-Form)

For $\beta = A\,dy \wedge dz + B\,dz \wedge dx + C\,dx \wedge dy$,

$$d\beta = \Big(\frac{\partial A}{\partial x} + \frac{\partial B}{\partial y} + \frac{\partial C}{\partial z}\Big)\,dx \wedge dy \wedge dz. \tag{13}$$

The coefficient is the divergence of $\mathbf{V} = (A, B, C)$. Geometrically, the net outward flux of $\mathbf{V}$ through a small parallelepiped with edges $\varepsilon\mathbf{u}, \varepsilon\mathbf{v}, \varepsilon\mathbf{w}$ (right-handed) is

$$\varepsilon^3\,(d\beta)(\mathbf{u}, \mathbf{v}, \mathbf{w}) + o(\varepsilon^3) = \varepsilon^3\,(\nabla \cdot \mathbf{V})\,\det(\mathbf{u}, \mathbf{v}, \mathbf{w}) + o(\varepsilon^3).$$

This is the same structure as in Section 5, one degree higher:

$$\text{circulation of a patch} = \text{rotation density} \times \text{oriented area},$$
$$\text{flux out of a small cube} = \text{flux density} \times \text{oriented volume}.$$

Example 6.4 (An Expanding Field)

For $\beta = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy$ (the radial field $\mathbf{V} = (x, y, z)$), $A = x$, $B = y$, $C = z$ and

$$d\beta = (1 + 1 + 1)\,dx \wedge dy \wedge dz = 3\,dx \wedge dy \wedge dz.$$

A small cube of side ε has a net outward flux of $3\varepsilon^3$: the field expands space uniformly, just as the radial field of Section 3 did not rotate anything but did expand it.